MTH 346/646 - Class 20 - 11/4

Fun fact: The proof of Fermat's last theorem:

Assume that there are positive integers a, b, c and $n \geq 3$ so that $a^n + b^n = c^n$.

Using these, create the elliptic curve

$$Y^2 = X(X - a^n)(X + b^n).$$

The discriminant of the cubic is $a^n b^n c^n$ is quite large, but the conductor (roughly the product of primes of bad reduction) is really small. This is a very weird property.

Andrew Wiles proves that this elliptic curve is so weird that it doesn't exist. Therefore Fermat's last theorem is true.

Request: Please let me know what your plans are about the two projects for this class. I have heard from 3 of the

Last time: We proved that if r is the rank, then $2^r = \frac{|\text{im } \alpha||\text{im } \overline{\alpha}|}{4}$. We also talked about the example of $y^2 = x^3 - 3x$.

Example 1: $C : y^2 = x^3 + 13x$, $\overline{C} : y^2 = x^3 - 52x$.

For α, we need to check the values in $\{\pm 1, \pm 13\}$. We cannot have -1 and -13 be in the image of α, because then we'd have solutions to $N^2 = -M^4 - 13e^4$ or $N^2 = -13M^4 - e^4$.

We have that 1 is in the image of α, because it is the image of $\mathcal{O}$ and 13 is in the image of α because $\alpha((0,0)) = 13$.

For $\overline{\alpha}$ we need to check $\{\pm 1, \pm 2, \pm 13, \pm 26\}$. Again, we know that $1 \in \text{im } \overline{\alpha}$, $-13 \in \text{im } \overline{\alpha}$ because $\overline{\alpha}(0,0) = -52 \equiv -13 \pmod{(\mathbb{Q}^\times)^2}$.

For -1, we are looking for solutions to

$$N^2 = -M^4 + 52e^4.$$

Observe that $52 - 16 = 36$. Thus, if we set $M = 2$ and $e = 1$, we get $6^2 = -(2)^4 + 52 \cdot 1^4$. Hence, there is a solution to this and so -1 is in the image of $\overline{\alpha}$. Since -13 is in the image, we know that 13 is in the image also (because the image is a subgroup).

For 2, we are looking for solutions to

$$N^2 = 2M^4 - 26e^4.$$

The right hand side is even, so N is even, and so $2M^4 - 26e^4$ is a multiple of 4 and so $M^4 - 13e^4$ is even. Since $\gcd(M, e) = 1$, this means that M and e are both odd. However, if $M = (2k+1)$, then

$$M^4 = 16k^4 + 32k^3 + 24k^2 + 8k + 1 = 16(k^4 + 2k^3 + \frac{k(3k+1)}{2}) + 1.$$

If k is even, then $k(3k+1)/2$ is an integer. If k is odd, then $3k+1$ is an integer and so $k(3k+1)/2$ is an integer. It follows that $M^4 \equiv 1 \pmod{16}$. Hence

$$N^2 \equiv 2 - 26 \equiv 8 \pmod{16}.$$

This shows that 2 is NOT in the image of α.

So we know that $\{1, -1, 13, -13\}$ are in the image, and 2 is not in the image. By closure, -2, 26, and -26 aren't in the image either.

Thus, we get that $|\text{im } \alpha| = 2$ and $|\text{im } \overline{\alpha}| = 4$, so $2^r = \frac{|\text{im } \alpha||\text{im } \overline{\alpha}|}{4} = 2$. Thus, $r = 1$, and the rank of C is 1.

How do we get a point of infinite order on C? The one "non-trivial" solution we found was with $-1 \in \text{im } \overline{\alpha}$. We had $M = 2$, $e = 1$ and $N = 6$. This gives us the point

$$\left(\frac{dM^2}{e^2}, \frac{dMN}{e^3} \right) = (-4, -12)$$

on $\overline{C} : y^2 = x^3 - 52x$. We can map this point back to C using ψ and get a point with x-coordinate $\frac{y^2}{4x^2} = \frac{144}{4 \cdot 4^2} = \frac{9}{4}$. This gives us the point $P = (9/4, 51/8)$. Would you have found this by searching?

The point $P = (9/4, 51/8)$ has infinite order in C. We don't know for sure if it is a generator of the Mordell-Weil group. (Magma tells us that it is.)

Interpretation:

For each divisor d of b, we have the curve

$$C_d : y^2 = dx^4 + ax^2 + (b/d)$$

and a map $f_d : C_d \to C$ given by $f(x, y) = (dx^2, dxy)$. Every rational point on C is the image of some rational point on C_d (for some d).

Each curve C_d has genus one, and if C_d has a rational point on it, then C_d is isomorphic to an elliptic curve. (The map f_d is never an isomorphism. If C_d has a rational point, then $C_d \cong \overline{C}$.)

Also, the points on C_d are simpler than the points on C, so searching for points on C_d is easier than searching on C.

For each squarefree $d|b$, we try to either

(a) find a rational point on C_d, (If this happens we know that d is in the image of α.)

(b) use modular arithmetic to prove there aren't any such points. (If this happens we know that d isn't in the image of α.)

The curves C_d are called *homogeneous spaces* or *two-covers* of C, and the procedure for computing the rank of C is called a two-descent.

Q: What if we can't find a rational point on C_d, and we can't use modular arithmetic to prove that C_d doesn't have any rational points????

Example 2: What's the rank of $C : y^2 = x^3 + 17x$.

The image of $\alpha \subseteq \{\pm 1, \pm 17\}$. We have $\alpha(\mathcal{O}) = 1$ and $\alpha(T) = 17$. It's easy to see that if $d < 0$, then d is not in the image of α and so $|\text{im } \alpha| = 2$.

Now, $\overline{C} : y^2 = x^3 - 52x$. The image of $\overline{\alpha} \subseteq \{\pm 1, \pm 2, \pm 17, \pm 34\}$. We have $\overline{\alpha}(\mathcal{O}) = 1$ and $\overline{\alpha}(\overline{T}) \equiv -52 \equiv -17 \pmod{(\mathbb{Q}^\times)^2}$. So we know that 1 and -17 are in the image. Is there anything else in the image? This is not easy to answer.

The reason is that for each of the six remaining choices of d, the curve $N^2 = dM^4 - (52/d)e^4$ has solutions mod n for every positive integer n (and the equation also has real solutions), but searching on them does not reveal points.

If there are no solutions, then $|\text{im } \overline{\alpha}| = 2$ and the rank of the curve is zero. But how do you prove that? How do you show that $2 \notin \text{im } \overline{\alpha}$? This means that

$$N^2 = 2M^4 - 34e^4$$

has no solutions.

It is possible to do a *second two-descent* and show that every point on the curve $y^2 = 2x^4 - 34$ comes from a point on either

$$\begin{aligned} 10A^2 + 34C^2 &= 17U^2 + V^2 \\ A^2 + 5C^2 &= UV \end{aligned}$$

or

$$\begin{aligned} 5A^2 + 17C^2 &= 17U^2 + V^2 \\ A^2 + 5C^2 &= 2UV. \end{aligned}$$

One can show that neither of these curves can have points on them mod 17.

Note: There are times when doing a second two-descent isn't successful and perhaps a third two-descent must be done, or perhaps a fourth. We have no guarantee that eventually we'll get an equation with a solution, or an equation that we can show has no solutions by looking at it mod some integer.

Example 3: Find all non-negative integers n for which

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = m^6$$

for some integer m. (I might ask a question like this on the final.)

The left hand side has a formula in terms of n, it is $\frac{n^2(n+1)^2}{4}$. Thus, we're really trying to solve $\frac{n(n+1)}{2} = m^3$.

Set $x = m$ and $y = n$. Then we get $x^3 = \frac{y(y+1)}{2}$. Multiply both sides by 8 and get $8x^3 = 4y(y+1)$. Adding one to both sides gives $8x^3 + 1 = 4y(y+1) + 1 = 4y^2 + 4y + 1 = (2y+1)^2$. Now, let $X = 2x$ and $Y = 2y + 1$. Then we get $Y^2 = X^3 + 1$.

One solution to this equation is $Y = 0$ and $X = -1$. This point has order 2. We can translate this point to $(0,0)$ by replacing X with $X - 1$. We get

$$C : Y^2 = (X-1)^3 + 1 = X^3 - 3X^2 + 3X.$$

We'll compute the rank of this curve.

For α we have im $\alpha \subseteq \{\pm 1, \pm 3\}$. We have $\alpha(\mathcal{O}) = 1$ and $\alpha(T) = 3$. If $d < 0$, this gives $N^2 = dM^4 - 3M^2e^2 + (3/d)e^4$ and the right hand side is < 0. Thus, im $\alpha = \{1, 3\}$.

The curve $\overline{C} : Y^2 = X^3 - 6X^2 + ((-3)^2 - 4 \cdot 3) = X^3 - 6X^2 - 3X$. Thus, the image of $\overline{\alpha} \subseteq \{\pm 1, \pm 3\}$. We have $\overline{\alpha}(\mathcal{O}) = 1$ and $\overline{\alpha}(\overline{T}) = -3$. If -1 is in the image of $\overline{\alpha}$, we'd have a solution to

$$N^2 = -M^4 - 6M^2e^2 + 3e^4.$$

Looking at this mod 3 gives $N^2 \equiv -M^4 \pmod 3$ and this gives $N \equiv M \equiv 0 \pmod 3$. This implies that $9|3e^4$ and this implies that $3|e$, which is a contradiction. Hence, -1 is not in the image of $\overline{\alpha}$. Since im $\overline{\alpha}$ is a subgroup, we have that 3 isn't in the image either (otherwise $3 * (-3) \equiv -1 \pmod{(\mathbb{Q}^\times)^2}$ would be).

It follows that $|\text{im } \alpha| = 2 = |\text{im } \overline{\alpha}|$ and so the rank r satisfies $2^r = \frac{|\text{im } \alpha||\text{im } \overline{\alpha}|}{4} = 4/4 = 1$ so $r = 0$.

Thus, this elliptic curve has only finitely many rational points on it, and the same is true with $Y^2 = X^3 + 1$, since it is isomorphic to C. The discriminant of $Y^2 = X^3 + 1$ is -27 and if (x, y) is a point of finite order on C we have that $Y = 0$ (and so $X = -1$) or $y^2| - 27$. This implies that $y = \pm 1$ or $y = \pm 3$. If $Y = \pm 1$, we get $X = 0$. If $Y = \pm 3$, we get $X = 2$. (We know that all of these points have finite order, because we've already computed the rank and found that there aren't *any* points of infinite order on the curve.)

Thus, the curve has exactly six rational points on it: $\mathcal{O}$, $(-1, 0)$, $(0, \pm 1)$ and $(2, \pm 3)$. The only points on this curve where Y is an odd integer > 0 are $(0, 1)$ and $(2, 3)$ and so $3 = 2n + 1$ (and $n = 1$) or $1 = 2n + 1$ (and $n = 0$).

Example 4: (With Magma) Let's see if we can use Magma to help us compute the rank of $C : y^2 = x^3 + 1918x$. The curve $\overline{C} : y^2 = x^3 - 7672x$.

The divisors of 1918 are $\{\pm 1, \pm 2, \pm 7, \pm 14, \pm 137, \pm 274, \pm 959, \pm 1918\}$. We can rule out the negative values because $N^2 = dM^4 + (1918/d)e^4$ isn't solvable if $d < 0$.

We have $\alpha(\mathcal{O}) = 1$ and $\alpha(T) = 1918$. For $d = 2$ we get

$$N^2 = 2M^4 + 959e^4.$$

Note that $961 = 31^2$ and so we can set $M = e = 1$ and $N = 31$. Thus 2 is in the image and hence $959 \equiv 2 \cdot 1918 \pmod{(\mathbb{Q}^\times)^2}$ is too. So we know that the image of α contains $\{1, 2, 959, 1918\}$. Since the image of α is a subgroup, it must either be all positive possibilities, or just these.

For $d = 7$ we get

$$N^2 = 7M^4 + 274e^4.$$

To deal with this we create the hyperelliptic curve $y^2 = 7x^4 + 274$ in Magma. (Here $y = N/e^2$, $x = M/e$.)

```
R<x> := PolynomialRing(Rationals());
```

```
C := HyperellipticCurve(7*x^4+274);
```

We search for points using Magma

```
Points(C : Bound := 100);
```

and find many. The tuples that are returned are $(M : N : e)$. One such is $M = 3$, $e = 1$ and $N = 29$. Thus, 7 is in the image and so $7 * 1918 = 274$ is in the image and so the image must have size at least 5. It follows from this that the image has size 8, so $|\text{im } \alpha| = 8$.

Now, for $\overline{C} : y^2 = x^3 - 7672x$, we have $\text{im } \overline{\alpha} \subseteq \{\pm 1, \pm 2, \pm 7, \pm 14, \pm 137, \pm 274, \pm 959, \pm 1918\}$ again. We know that $\overline{\alpha}(\mathcal{O}) = 1$ and $\overline{\alpha}(\overline{T}) = -7672 \equiv -1918$.

For $d = -1$ we get $N^2 = -M^4 + 7672e^4$. We search for points on the curve $y^2 = -x^4 + 7672$ and don't find any.

```
R<x> := PolynomialRing(Rationals());
```

```
C := HyperellipticCurve(-x^4+7672);
```

```
Points(C : Bound := 100);
```

The command `IsLocallySolvable(C,2);` returns false. This means that there aren't solutions to $N^2 = -M^4 + 7672e^4$ modulo high powers of 2. So -1 isn't in the image. This shows that 1918 isn't in the image either.

For $d = 2$ we get $N^2 = 2M^4 - 3836e^4$ and Magma finds points on this curve, including $M = 8$, $e = 1$ and $N = 66$. Thus, 2 is in the image, and so $2 * (-1918) \equiv -959$ is in the image too.

We can conclude that -2 is not in the image, since otherwise $-1 \equiv 2 * (-2)$ would be. Also, 959 isn't in the image since $2 \equiv 959 \equiv 1918$.

For $d = 7$ we get $N^2 = 7M^4 - 1096e^4$. Again, this curve has no solutions modulo high powers of 2, so 7 is not in the image. This shows that 14, -137 and -274 aren't in the image either.

For $d = -7$ we get $y^2 = -7x^4 + 1096$. Setting $x = 1$ we get $1089 = 33^2$, so -7 is in the image. Thus, -14, 137 and 274 are in the image.

In summary, im $\overline{\alpha} = \{1, 2, -7, -14, 137, 274, -959, -1918\}$. Thus, we have

$$2^r = \frac{|\text{im } \alpha||\text{im } \overline{\alpha}|}{4} = \frac{8 \cdot 8}{4} = 16.$$

Thus, $r = 4$.